

ThetaTrap

A bounded MCP-native AI options agent

PAPER TRADING / BASIC INDICATIVE DATA



WHY IT EXISTS

I started with a video-inspired question: can an agent collect elevated earnings option premium without making a directional bet? The strategy was easier than the automation. Four-leg prices can disagree, events can be ambiguous, model output can drift, and a timeout cannot be allowed to create a duplicate order.

STRATEGY IN PLAIN ENGLISH

ThetaTrap freezes first-party-verified after-close earnings events, compares near- and next-week implied volatility, checks quote freshness and liquidity, and constructs one symmetric iron condor. Selling the inner put and call spreads collects credit; the farther-out wings bound the maximum loss. The target lifecycle is one afternoon entry and a complete next-morning exit.

AGENT BOUNDARY

Featherless-hosted Qwen calls approved tools, examines account, market, position, order and news evidence, and either vetoes the candidate or issues the exact pre-authorized Alpaca MCP order call. Deterministic Python owns the symbols, expirations, strikes, price, quantity, risk checks, one-shot authorization, reconciliation and exit.

WHY MCP MATTERS

Every Alpaca read and mutation crosses the pinned official MCP server. There is no direct SDK or REST fallback in application code, so the same typed, logged tool boundary judges can inspect is the boundary used by the agent.

\$500

MAX LOSS

1

CONTRACT

1

ENTRY / DATE

FAIL-CLOSED CONTROLS

- Maximum loss is also capped at 0.5% of observed equity.
- One position at a time and one broker submission per strategy date.
- Deterministic client IDs and broker reconciliation handle timeouts and restarts.
- The kill switch blocks new exposure but never blocks position-reducing exits.
- Assignment, broken legs or uncertain state require manual broker review.

RELEASE EVIDENCE

- 181 automated tests at the submission checkpoint.
- Five broker-isolated replay scenarios with zero external mutations.
- Seven eligible events and two explicit, source-backed exclusions.
- Persistent SQLite audit from model decision through account equity.
- Separate public read-only and private operator dashboards.

CURRENT RESULT + LIMITS

The official paper result is reported only after Alpaca confirms zero positions and zero open orders. Until then, no replay or simulated fill is presented as profit. Basic indicative quotes are not consolidated OPRA data, paper fills do not reproduce live execution, and one competition week cannot establish long-run expected value.